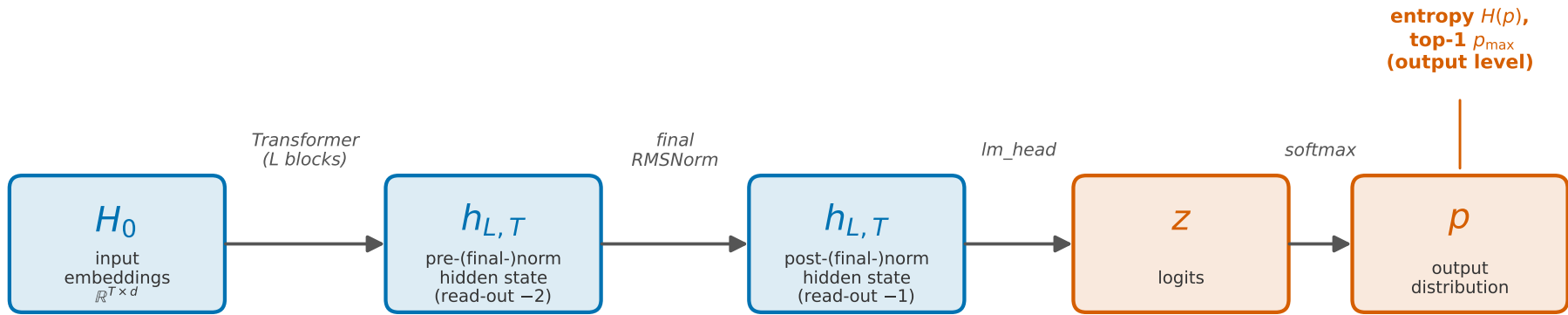




$$J = \partial h_{L,T} / \partial H_0 : \sigma_{\max}, \|V\|_F / \lambda_{\text{bulk}}, \text{ stable rank}$$

Algorithm 2 (pre-norm) + Algorithm 1 sampled-direction $\lambda_L^{(\varepsilon)}$

$$\partial h_{L,T}^{\text{post}} / \partial H_0 : \sigma_{\max} \text{ (post-norm)}$$

Algorithm 2 (post-norm) — the final normalization changes the hidden metric (App. C)

χ_F : Fisher/KL output susceptibility

how an input perturbation moves the output distribution p

$\nabla_{H_0} H$: entropy input-gradient (one backward)

the like-for-like partner of $\sigma_{\max}$ — the corrected comparison (Sec. 4.4)